Map building

Stored map:
embeddings, features, 3D points
+ keyframe poses

Known training trajectory
+ camera calibration

Training frames

Keyframe selection

DINOv2
Global embeddings

ALIKED
Local features

LightGlue
Training-pair matching

Triangulation
+ geometric checks

Features linked to 3D

Save

Save

Training embeddings
Image similarity

Keyframe features
Keypoints + descriptors

3D landmarks
Point coordinates

Load

Load

Retrieval index
Embeddings + poses

GPU feature cache
Upload on first use

3D points

Inference

DINOv2
Resize + normalize

Query embedding

Top-6 keyframes
Per-camera prior

Selected keyframes

New frames
Decode + camera crop

ALIKED
Query features

Query features

LightGlue
Match candidates

2D ↔ 3D matches

PnP + quality gates
2D ↔ 3D → pose

Initial pose

Visual pose

Initialization

Borders and poles
Map alignment

Visual corrections

EKF
Sensor fusion

Vehicle pose
x, y, yaw

Telemetry
Velocities + yaw rate

Motion prediction

Prediction

Embeddings retrieve similar views. Local features match image points. 3D landmarks turn those matches into a metric vehicle pose.